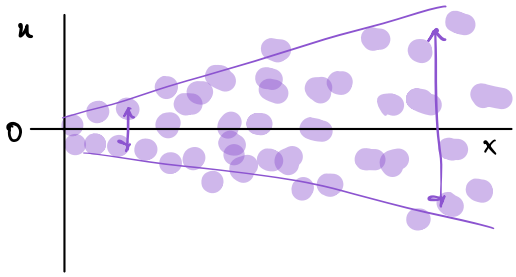
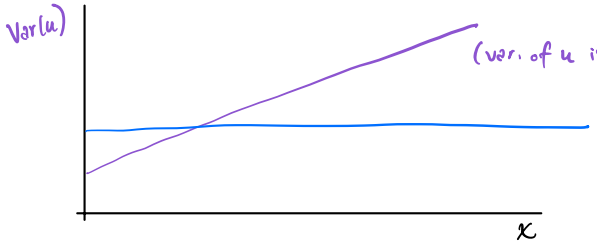
Graphs

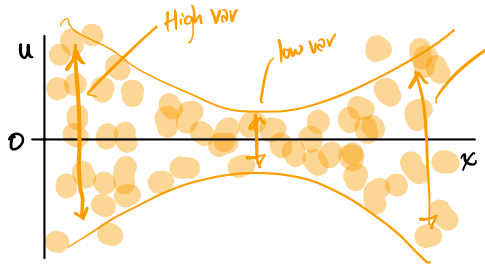


Var(u) is increasing w/. x
⇒ violation of homosk.
(we have heterosk.)
↳ OLS is still unbiased for estimating β coefs
... but it is biased for SEs & inefficient

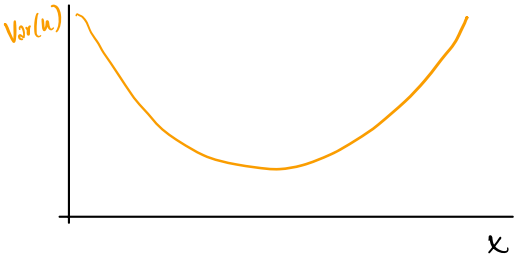


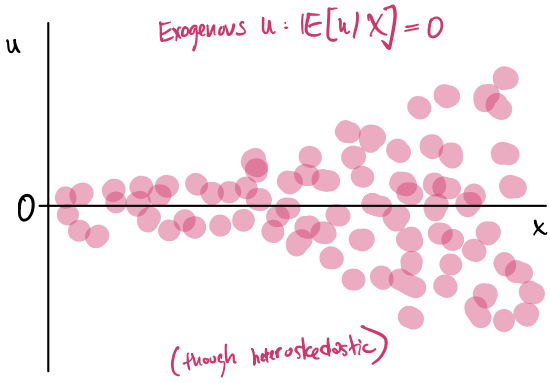
(var. of u is inc. w/. x)

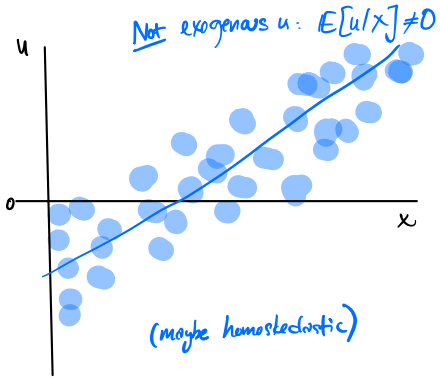
Homosk. should be constant variance across x



Again, heteroskedastic dist. seems very likely b/c the var(u) is not constant...







White test for heterosk.

1. Est. your regression of interest, eg,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

$$\rightarrow \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$

2. Get residuals, $y_i - \hat{y}_i = e_i$

NOTE: We are going to use e_i^2 to learn about u_i^2 which tells us about $Var(u_i) = \sigma_i^2$

3. Est. White regression

$$e^2 = \alpha_0 + \boxed{\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 +} \quad \text{original terms ("funnel" situation)}$$

$$+ \boxed{\alpha_4 x_1 x_2 + \alpha_5 x_1 x_3 + \alpha_6 x_2 x_3} + \quad \text{two-way interactions}$$

$$+ \boxed{\alpha_7 x_1^2 + \alpha_8 x_2^2 + \alpha_9 x_3^2} + v \quad \text{squared terms}$$

These lines are where White "beats" G-Q

4. Calc. the White test stat

R^2 from regression above

$$LM = \underset{\#\text{ obs}}{n} \times R_e^2$$

Test against χ^2_k where k = # of coeficients in White reg.

(Classical) Measurement error

We want to know a specific variable z but we actually observe

$$x = z + \varepsilon \quad (\text{and } \varepsilon \text{ is "well behaved"})$$

If we regress $y \sim x$ then $\text{plim}\ \hat{\beta} = \dfrac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2}\beta$

If we regress mismeasured y on a well-measured variable

$$\tilde{y} = y + \varepsilon \qquad \tilde{y} = \beta_0 + \beta_1 x + u$$

$$y + \varepsilon = \beta_0 + \beta_1 x + u$$

$$y = \beta_0 + \beta_1 x + (u - \varepsilon)$$

Classical meas. error in y:

1. No bias in $\hat{\beta}$
2. Bigger SEs